Wenyu Du
https://wenyudu.github.io/

Email : wenyudu@yahoo.com
Mobile : +852-6740-1496

Professional Experience

- **DualverseAI** (Visit: dualverse.ai) Nov. 2025 – Present
  Co-Founder *Hong Kong*
  - Co-founded the company to build **"The Station"**, a generative simulation platform for AI-driven scientific discovery.
  - Building open-world environments where AI agents perform autonomous reasoning and experimentation.
  - Leveraging the simulation environment as a dynamic testbed to explore mechanistic interpretability and model reasoning.

Education

- **University of Hong Kong** Mar. 2021 – Nov. 2025
  Doctor of Philosophy in Computer Science
- **University of Southampton** Sep. 2016 – Nov. 2017
  Master of Science in Data Science with Distinction
- **University of Nottingham** Sep. 2011 – July. 2016 (Sep. 2013 – Sep. 2014, Sick leave)
  Bachelor of Science in Computer Science with High 2:1

Selected Publications

- **Wenyu Du**, Stephen Chung. *"Can AI Agents Make Open-Ended Scientific Discovery? Evidence from Station"*, arXiv preprint arXiv:2610.08927, 2026.
- Stephen Chung, **Wenyu Du**, William J. Wesley. *"Autonomous Mathematical Discovery in an Open-World Multi-Agent Environment"*, arXiv preprint arXiv:2608.23691, 2026.
- Jiaxi Bi*, Tongxu Luo*, **Wenyu Du**, Zhengyang Tang, Benyou Wang. *"Cut Your Losses! Learning to Prune Paths Early for Efficient Parallel Reasoning"*, (ACL, Oral), 2026.
- Stephen Chung, **Wenyu Du**. *"The Station: An Open-World Environment for AI-Driven Discovery"*, Preprint, 2025.
- Stephen Chung*, **Wenyu Du***, Jie Fu. *"Thinker: Learning to Think Fast and Slow"*, Accepted to The Thirty-Nineth Annual Conference on Neural Information Processing Systems (NeurIPS, Poster), 2025.
- Yifan Zhang, **Wenyu Du**, Dongming Jin, Jie Fu, Zhi Jin. *"Finite State Automata Inside Transformers with Chain-of-Thought: A Mechanistic Study on State Tracking"*, (ACL, Oral), 2025.
- **Wenyu Du***, Tongxu Luo*, Zihan Qiu, Zeyu Huang, Yikang Shen, Reynold Cheng, Yike Guo, Jie Fu, *"Stacking Your Transformers: A Closer Look at Model Growth for Efficient LLM Pre-Training"*, Accepted to The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS, Spotlight), 2024.
- **Wenyu Du***, Shuang Cheng*, Tongxu Luo, Zihan Qiu, Zeyu Huang, Ka Chun Cheung, Reynold Cheng, Jie Fu, *" Unlocking Continual Learning Abilities in Language Models"*, Accepted to The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP, Findings), 2024.
- Jacob Louis Hoover, **Wenyu Du**, Alessandro Sordoni, Timothy J. O'Donnell, *"Linguistic Dependencies and Statistical Dependence"*, The 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP, Main), 2021.

- **Wenyu Du***, Zhouhan Lin*, Yikang Shen, Timothy J. O'Donnell, Yoshua Bengio, Yue Zhang, *"Exploiting Syntactic Structure for Better Language Modeling: A Syntactic Distance Approach"*, The 58th Annual Meeting of the Association for Computational Linguistics (ACL, Main), 2020.

  * = equal contribution.

Publications

- **Wenyu Du**, Stephen Chung. *"Can AI Agents Make Open-Ended Scientific Discovery? Evidence from Station"*, arXiv preprint arXiv:2610.08927, 2026.
- Stephen Chung, **Wenyu Du**, William J. Wesley. *"Autonomous Mathematical Discovery in an Open-World Multi-Agent Environment"*, arXiv preprint arXiv:2608.23691, 2026.
- Jiaxi Bi*, Tongxu Luo*, **Wenyu Du**, Zhengyang Tang, Benyou Wang. *"Cut Your Losses! Learning to Prune Paths Early for Efficient Parallel Reasoning"*, (ACL, Oral), 2026.
- Stephen Chung, **Wenyu Du**. *"The Station: An Open-World Environment for AI-Driven Discovery"*, Preprint, 2025.
- **Wenyu Du**, Rui Zheng, Tongxu Luo, Stephen Chung, Jie Fu *"Fine-Grained Manipulation of Arithmetic Neurons"*, Accepted to The 2025 Conference on Empirical Methods in Natural Language Processing (EMNLP, BlackboxNLP Workshop), 2025.
- Stephen Chung*, **Wenyu Du***, Jie Fu. *"Thinker: Learning to Think Fast and Slow"*, Accepted to The Thirty-Nineth Annual Conference on Neural Information Processing Systems (NeurIPS, Poster), 2025.
- Tongxu Luo*, **Wenyu Du***, Jiaxi Bi, Stephen Chung, Zhengyang Tang, Hao Yang, Min Zhang, Benyou Wang. *"Learning from Peers in Reasoning Models"*, Preprint, 2025.
- Stephen Chung*, **Wenyu Du***, Jie Fu. *"Learning from Failures in Multi-Attempt Reinforcement Learning"*, Preprint, 2025.
- Tongxu Luo*, **Wenyu Du***, Hanwen Hao, Min Zhang, Hao Yang, Benyou Wang. *"E2-RAG: Towards Editable Efficient RAG by Editing Compressed KV Caches"*, COLM, 2025.
- Xu Wang, Yan Hu, **Wenyu Du**, Reynold Cheng, Benyou Wang, Difan Zou. *"Towards Understanding Fine-Tuning Mechanisms of LLMs via Circuit Analysis"*, Accepted to The Forty-Second International Conference on Machine Learning (ICML, Poster), 2025.
- Yifan Zhang, **Wenyu Du**, Dongming Jin, Jie Fu, Zhi Jin. *"Finite State Automata Inside Transformers with Chain-of-Thought: A Mechanistic Study on State Tracking"*, (ACL, Oral), 2025.
- **Wenyu Du***, Tongxu Luo*, Zihan Qiu, Zeyu Huang, Yikang Shen, Reynold Cheng, Yike Guo, Jie Fu, *"Stacking Your Transformers: A Closer Look at Model Growth for Efficient LLM Pre-Training"*, Accepted to The Thirty-Eighth Annual Conference on Neural Information Processing Systems (NeurIPS, Spotlight), 2024.
- **Wenyu Du***, Shuang Cheng*, Tongxu Luo, Zihan Qiu, Zeyu Huang, Ka Chun Cheung, Reynold Cheng, Jie Fu, *" Unlocking Continual Learning Abilities in Language Models"*, Accepted to The 2024 Conference on Empirical Methods in Natural Language Processing (EMNLP, Findings), 2024.
- Ka Man Lo, Yiming Liang, **Wenyu Du**, Yuantao Fan, Zili Wang, Wenhao Huang, Lei Ma, Jie Fu, *"m2mKD: Module-to-Module Knowledge Distillation for Modular Transformers"*, arXiv:2402.16918.

- Jinyang Li, Binyuan Hui, Reynold Cheng, Bowen Qin, Chenhao Ma, Nan Huo, Fei Huang, **Wenyu Du**, Luo Si, Yongbin Li, *"Graphix-T5: Mixing Pre-Trained Transformers with Graph-Aware Layers for Text-to-SQL Parsing"*, Accepted to The 37th AAAI Conference on Artificial Intelligence (AAAI, Oral) 2023.
- Yuqiao Wen, Zichao Li, **Wenyu Du**, Lili Mou, *" f-Divergence Minimization for Sequence-Level Knowledge Distillation"*, The 61st Annual Meeting of the Association for Computational Linguistics (ACL, Main), 2023.
- Jacob Louis Hoover, **Wenyu Du**, Alessandro Sordoni, Timothy J. O'Donnell, *"Linguistic Dependencies and Statistical Dependence"*, The 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP, Main), 2021.
- Qiankun Fu, Linfeng Song, **Wenyu Du**, Yue Zhang, *"End-to-End AMR Corefencence Resolution"*, The 59th Annual Meeting of the Association for Computational Linguistics (ACL, Main), 2021.
- **Wenyu Du**[*], Zhouhan Lin[*], Yikang Shen, Timothy J. O'Donnell, Yoshua Bengio, Yue Zhang, *"Exploiting Syntactic Structure for Better Language Modeling: A Syntactic Distance Approach"*, The 58th Annual Meeting of the Association for Computational Linguistics (ACL, Main), 2020.
- Sushen Zhang, Ruijuan Chen, **Wenyu Du**, Ye Yuan, Vassilios S. Vassiliadis, *"A Hessian-Free Gradient Flow (HFGF) Method for the Optimisation of Deep Learning Neural Networks"*, Computers and Chemical Engineering (Journal), 2020.
- **Wenyu Du**, Baocheng Li, Min Yang, Qiang Qu, *"A Multi-Task Learning Approach for Answer Selection: A Study and a Chinese Law Dataset"*, The 33rd AAAI Conference on Artificial Intelligence (AAAI, student abstract), 2019.
- **Wenyu Du**, Shuai Yu, Min Yang, Qiang Qu, Jia Zhu, *"GPSP: Graph Partition and Space Projection based Approach for Heterogeneous Network Embedding"*, The 27th International World Wide Web Conference (WWW, poster), 2018.

[*] = equal contribution.

Selected Research Experience

- **NVIDIA** Jun. 2024 - Aug. 2024
  Research Intern
  - Project: Unlocking continual learning ability in language models, accepted by EMNLP 2024
- **Westlake University** Sep. 2018 - Feb. 2021
  Research Assistant
  - Project: Exploiting Syntactic Structure for Better Language Modeling: A Syntactic Distance Approach, accepted by ACL 2020

Programming Languages

- **Deep Learning**: Tensorflow, Pytorch
- **Languages/Skills**: Python, Java, MATLAB, C, C++

Language Certificate

- **IELTS:** Overall: 7.5 (Listening: 8.0, Speaking: 6.5, Reading: 8.5, Writing: 6.5)